Mithi River Water Quality Deep Learning Comparison Report

Classification and Regression Model Performance Analysis

Window size: 6 | Batch size: 64 | Epochs: 12 | Learning rate: 0.001

Classification Performance Comparison

Model	Accuracy	Macro F1	Weighted F1
ANN	0.9582	0.6718	0.9663
CNN	0.9722	0.7688	0.9759
RNN	0.9680	0.7239	0.9730
LSTM	0.9740	0.7570	0.9772

Best classification model: CNN (selected using macro-F1 to handle class imbalance fairly, with strong recall on poor/unsafe conditions).

Regression Performance Comparison

Model	MAE	RMSE	R2 Score
ANN	3.7264585	6.1118836	0.6289776
CNN	25.018335	38.105656	0.27110282
RNN	781.3628	1556.316	-0.85633516
LSTM	781.16327	1555.9991	-0.8554211

Best regression model: ANN (selected using lowest RMSE and highest R2 score for predicting TDS, BOD, and COD).

Key Insight

The data is highly imbalanced toward the Poor class, so macro-F1 was used for model selection instead of accuracy alone. This prevents the model from appearing strong simply by predicting the majority class. The best classification model was the CNN, while the best regression model was the ANN.